

SmartPantry

Project Requirements Specification Document

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Introduction

SmartPantry is a pantry-aware recommendation and decision-support system designed to help households reduce food waste and simplify meal planning. The application combines pantry inventory tracking, expiration analysis, recipe matching, recommendation scoring, grocery planning, and analytics-driven dashboards to generate practical meal recommendations based on ingredients users already have available at home. Unlike traditional recipe applications that primarily focus on recipe discovery, SmartPantry prioritizes inventory awareness, expiration urgency, missing ingredient count, cooking time, and user preferences to create more realistic and efficient meal recommendations. The goal of the project is to improve household meal planning efficiency while helping users reduce avoidable food waste and unnecessary grocery purchases.

Project Objectives

The primary objectives of SmartPantry are:

- Reduce household food waste through expiration-aware meal recommendations and improved pantry visibility.
- Simplify meal planning by generating practical recommendations based on ingredients users already have available at home.
- Improve recommendation quality by combining ingredient availability, expiration urgency, cooking time, missing ingredient count, and user preferences into a multi-factor recommendation system.
- Help users reduce unnecessary grocery spending by prioritizing meals that require fewer additional ingredients.
- Develop a pantry inventory management system that allows users to organize, update, and monitor household food inventory efficiently.
- Provide analytics and reporting features that help users understand pantry usage behavior, expiration trends, and food waste patterns over time.
- Create a user-friendly and scalable software architecture that balances recommendation accuracy, usability, and long-term maintainability.
- Evaluate whether a pantry-aware recommendation system can improve household meal planning efficiency compared with traditional manual meal planning methods.

Competitors & Market Trends

Meal planning applications and grocery management platforms have continued to grow in popularity as users search for faster and more practical ways to manage household food planning. Many existing applications focus heavily on recipe discovery, personalized meal inspiration, or grocery delivery integration. However, competitor research revealed that many users become frustrated when applications require too much manual inventory entry or when recommendations fail to account for ingredients users already own.

Another major trend identified during research is the increasing interest in recommendation systems that prioritize convenience, efficiency, and reduced waste. Many households purchase groceries with good intentions but fail to use ingredients before expiration because they struggle to decide what meals can realistically be prepared using available inventory. Existing systems often prioritize recipe quantity instead of recommendation practicality.

SmartPantry addresses these issues by focusing heavily on inventory visibility, expiration awareness, recommendation practicality, and simplified meal planning workflows.

Competitors and Market Research

Competitor Name	Sentiment	Feedback
Mealime	Positive	"The meal planning process feels simple and fast."
Mealime	Negative	"The recipe suggestions started feeling repetitive after a while."
Paprika Recipe Manager	Positive	"Inventory organization is easy to manage."
Paprika Recipe Manager	Negative	"Manual pantry entry takes too much time."
Samsung Food	Positive	"Recipe discovery and meal inspiration are very strong."

Competitor Name	Sentiment	Feedback
Samsung Food	Negative	"The interface feels cluttered because there are too many features."
SuperCook	Positive	"Ingredient matching works surprisingly well."
SuperCook	Negative	"The recommendations do not account for expiration dates."
AnyList	Positive	"The grocery list integration is very useful."
AnyList	Negative	"Meal planning and pantry tracking feel disconnected."
Yummly	Positive	"Personalized recommendations are helpful."
Yummly	Negative	"The application focuses more on browsing recipes than reducing waste."

Overall Positive Trends

1. Users prefer simple and fast meal planning workflows.
2. Ingredient-based recipe matching improves user engagement.
3. Grocery integration increases practical value for households.
4. Personalized recommendations improve user interaction.
5. Dashboard-style interfaces improve usability.

Overall Negative Trends

1. Manual pantry entry creates frustration for long-term use.
2. Recipe recommendations can become repetitive over time.
3. Existing applications often prioritize recipe browsing over pantry awareness.
4. Many systems ignore expiration urgency.
5. Some applications become cluttered with too many disconnected features

Stakeholders

The primary stakeholders for SmartPantry include end users and households, students and budget-conscious users, advisor and capstone committee members, usability testing participants, and the project developer. Each stakeholder group influences the project differently based on usability expectations, technical requirements, scalability concerns, and overall system functionality.

The highest priority stakeholders are the end users because the success of SmartPantry depends heavily on recommendation quality, usability, and inventory management efficiency. The advisor and capstone committee are also considered high-priority stakeholders because they are responsible for evaluating the technical depth, software engineering practices, scalability, and overall feasibility of the project. Research participants and usability testers are medium-priority stakeholders because they provide valuable feedback related to user experience, recommendation usefulness, interface design, and overall practicality.

The project developer is also considered a stakeholder because development scope, maintainability, architecture, and technical implementation directly affect project success. Future households and collaborative users are considered lower-priority stakeholders during the initial MVP phase because advanced collaboration systems are outside the scope of the first development cycle.

Stakeholder Needs

Stakeholder	Priority	Needs
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End Users/Households	High	Want fast and practical meal recommendations based on available inventory, expiration urgency, and household preferences.
Advisor/Capstone Committee	High	Want realistic project scope, technical depth, measurable evaluation metrics, proper software engineering practices, and defensible research outcomes.
Research Participants/ Test User	Medium	Want an interface that is understandable, responsive, and easy to navigate during usability testing.
Project Developer	Medium	Needs scalable architecture, maintainable recommendation logic, manageable development scope, and efficient database organization.
Budget- Conscious Users	Medium	Want recommendations that minimize missing ingredients and reduce unnecessary grocery spending.
Future Collaborative Households	Low	May benefit from future multi-user inventory systems and collaborative pantry management features.

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Target Audience/End User

The primary target audience for SmartPantry includes busy individuals, students, couples, and families who want to reduce food waste and simplify meal planning. These users may struggle with forgotten groceries, expiration management, or deciding what meals can realistically be prepared using ingredients already available at home.

A secondary audience includes budget-conscious households interested in improving grocery efficiency, reducing unnecessary purchases, and managing household inventory more effectively. Different user groups may interact with the system differently. Some users may primarily use the pantry inventory and expiration tracking features, while others may rely more heavily on recommendation systems, grocery list generation, and meal planning analytics. The application is specifically designed for users who want practical recommendations instead of simply browsing large recipe collections. SmartPantry focuses on helping users make faster household decisions using ingredients they already own rather than encouraging unnecessary grocery purchases or excessive recipe searching.

User Stories

The following user stories were developed to better understand how different users may interact with SmartPantry and what goals they are attempting to accomplish while using the system. These stories focus heavily on meal planning efficiency, inventory awareness, recommendation practicality, and reducing household food waste.

1. As a busy household user, I want SmartPantry to prioritize recipes using ingredients close to expiration so I can reduce food waste and avoid unnecessary grocery purchases.
2. As a user, I want to quickly enter pantry and refrigerator items so inventory tracking does not become frustrating or time-consuming.
3. As a user, I want recipe recommendations based on ingredients I already own so I can reduce additional grocery spending and make faster meal decisions.
4. As a user with limited cooking time, I want cooking-time filters so I can find meals that fit my daily schedule.
5. As a budget-conscious user, I want the system to recommend meals requiring fewer missing ingredients so I can minimize grocery costs.
6. As a user, I want expiration warnings for pantry items so I can use food before it spoils and reduce avoidable waste.

7. As a user, I want SmartPantry to generate grocery lists from selected meals so meal planning and grocery shopping feel connected and organized.
8. As a user, I want to review pantry usage and waste analytics so I can better understand my household food habits and improve grocery efficiency over time.
9. As a returning user, I want SmartPantry to remember my food preferences and previously selected meals so future recommendations become more personalized and practical.
10. As a user, I want meal recommendations to balance ingredient availability, expiration urgency, and cooking time so the suggestions feel realistic for my household situation.

Use Cases

Use cases are detailed scenarios that explain how users interact with SmartPantry to complete specific tasks. These use cases help define system behavior, user interaction flow, validation requirements, and expected outcomes within the application.

Use Case 1 – Generate Meal Recommendations

User Type

Household User

Goal of User

The user wants to generate practical meal recommendations using ingredients already available in the pantry, refrigerator, or freezer.

Preconditions

- User account exists within the system.
- Pantry inventory contains stored ingredients.

- Recipe dataset is available.
- Recommendation engine is active.

Triggers

- User selects the “Generate Recommendations” button from the dashboard.

Post-Conditions

- Ranked meal recommendations are displayed.
- Recommendation scores are calculated and stored.
- User may save meals or generate grocery lists.

Scenario

1. User logs into SmartPantry.
2. User opens the recommendation dashboard.
3. System retrieves pantry inventory data.
4. System retrieves available recipe dataset.
5. System compares pantry ingredients against recipe ingredients.
6. System calculates ingredient match percentage.
7. System calculates expiration urgency values.
8. System applies weighted recommendation scoring logic.
9. Recipes are ranked from highest to lowest recommendation score.
10. Ranked recommendations are displayed to the user.
11. User reviews recommendation explanations.
12. User selects a recommended meal.
13. User may save the recipe or generate a grocery list.

Exceptions

- Pantry inventory may be empty.
- Ingredient names may not perfectly match recipe dataset values.
- Recommendation results may be limited if inventory contains very few ingredients.
- Recommendation engine may return low-quality matches if recipe dataset is incomplete.

Use Case 2 – Update Pantry Inventory

User Type

Household User

Goal of User

The user wants to add, edit, or remove pantry items efficiently.

Preconditions

- User account exists within the system.
- Pantry database connection is active.

Triggers

- User opens pantry inventory management screen.

Post-Conditions

- Pantry inventory database is updated successfully.
- Dashboard refreshes updated inventory information.

Scenario

1. User logs into SmartPantry.
2. User selects pantry inventory page.
3. User chooses to add, edit, or remove pantry items.
4. User enters pantry item name.
5. User enters quantity information.
6. User enters expiration date if applicable.
7. System validates item information.
8. System updates inventory database.
9. Pantry dashboard refreshes.
10. Updated pantry inventory becomes available to recommendation engine.

Exceptions

- Duplicate ingredient names may exist.
- Invalid expiration dates may be entered.
- User may enter incomplete item information.
- Database update may fail if connection is interrupted.

Use Case 3 – Generate Grocery List**User Type**

Household User

Goal of User

The user wants to generate grocery lists from selected meal recommendations.

Preconditions

- Recommended meals exist within the system.
- Pantry inventory database is available.

Triggers

- User selects a recipe and requests grocery list generation.

Post-Conditions

- Grocery list is generated and displayed.
- Missing ingredients are identified and organized.

Scenario

1. User selects recommended meal.
2. System retrieves recipe ingredient list.
3. System compares recipe ingredients against pantry inventory.
4. Missing ingredients are identified.
5. Existing ingredients are removed from grocery calculation.
6. Grocery list is generated.
7. Grocery list is displayed to user.
8. User may export or save grocery list.

Exceptions

- Pantry quantities may be inaccurate.
- Ingredient normalization may fail for certain ingredient names.
- Recipe ingredient list may contain incomplete ingredient information.

Use Case 4 – Review Pantry Analytics

User Type

Household User

Goal of User

The user wants to review pantry usage behavior and food waste analytics.

Preconditions

- Pantry inventory history exists.
- Analytics data has been collected.

Triggers

- User selects analytics dashboard.

Post-Conditions

- Analytics charts and usage reports are displayed.

Scenario

1. User opens analytics dashboard.
2. System retrieves pantry usage history.
3. System retrieves expiration and waste data.
4. Analytics calculations are performed.
5. Charts and metrics are generated.
6. User reviews pantry usage trends.
7. User reviews expired item statistics.
8. User reviews recommendation usage behavior.

Exceptions

- Analytics data may be limited for new users.
- Pantry usage history may be incomplete.
- Visualizations may not load if analytics calculations fail.

Define Scope & Features

SmartPantry will consist of multiple integrated systems focused on recommendation generation, pantry inventory management, expiration tracking, grocery planning, and food waste analytics. The application is intentionally designed to balance technical depth with realistic development expectations for a graduate-level capstone project completed within a four-month development window.

The MVP version of SmartPantry will focus primarily on recommendation systems, pantry inventory management, expiration analysis, and analytics-driven meal planning workflows. Additional advanced features such as barcode scanning, OCR receipt scanning, household collaboration, and nutrition tracking may be considered future expansion features but are intentionally excluded from the initial MVP scope to maintain realistic development expectations.

The system combines recommendation algorithms, analytics processing, inventory management, database architecture, and user-centered interface design into one connected software platform.

- **Major Feature 1 – Pantry Inventory and Expiration Management System**

This feature allows users to add, edit, categorize, store, and manage pantry, refrigerator, and freezer inventory items. The system also calculates expiration urgency and flags ingredients that should be prioritized before spoilage occurs.

This feature combines:

- inventory management
- expiration analysis
- database persistence
- dashboard integration
- inventory categorization systems

The feature requires multiple connected systems and custom logic to maintain pantry tracking accuracy and expiration-aware recommendation functionality.

- **Major Feature 2 – Multi-Factor Recommendation Engine**

This feature analyzes pantry inventory, ingredient matching, expiration urgency, cooking time, missing ingredients, and user preferences to generate ranked meal recommendations.

The recommendation engine combines:

- weighted recommendation scoring
- recipe ranking algorithms
- ingredient normalization
- filtering systems
- recommendation analytics

This is the core technical feature of the project and represents the primary research and software engineering focus of SmartPantry.

- **Major Feature 3 – Meal Planning and Waste Analytics System**

This feature connects recipe recommendations, grocery list generation, pantry usage tracking, and food waste analytics into a unified planning dashboard.

The system allows users to:

- review pantry usage behavior
- identify food waste patterns
- track expired items
- monitor recommendation usage
- generate grocery planning reports

This feature combines analytics processing, visualization systems, inventory history tracking, and reporting functionality.

- **Minor Features**

1. Pantry category labels
2. Grocery list export
3. Favorite meals system
4. Cooking-time filters
5. Expiration notifications
6. Recipe detail pages
7. User preference settings
8. Basic analytics charts
9. Search and filtering functionality
10. Saved meal history

System Requirements

Requirement	Explanation	Solution
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SR1 – Pantry Inventory Database	Pantry inventory must be stored persistently for recommendation generation and analytics tracking.	SQLite or PostgreSQL
SR2 – Recommendation Scoring Engine	The system must calculate weighted recommendation scores using multiple recommendation factors.	Python recommendation algorithms
SR3 – Expiration Tracking System	The system must identify ingredients approaching expiration dates.	Python date-processing logic
SR4 – Recipe Matching System	Pantry ingredients must be compared against recipe datasets accurately.	Ingredient normalization algorithms
SR5 – Grocery List Generator	The system must generate grocery lists from selected meal recommendations.	Inventory comparison logic
SR6 – Waste Analytics Dashboard	Pantry usage and expiration analytics must be tracked visually.	Pandas analytics and visualization libraries
SR7 – User Preference Management	The system must store user dietary and recommendation preferences.	Relational database architecture

SR8 – Recommendation Filtering System	Users must filter recommendations based on cooking time and ingredient availability.	Python filtering logic
SR9 – Pantry Search Functionality	Users must search inventory items quickly and efficiently.	Search query filtering system
SR10 – Meal History Tracking	The system must store previously selected meals and recommendations.	Database history tables

Interface Requirements

Requirement	Explanation	Solution
IR1 – Pantry Dashboard	Users must view pantry inventory clearly and efficiently.	Streamlit or Flask dashboard interface
IR2 – Recommendation Dashboard	Users must view ranked meal recommendations and recommendation explanations.	Interactive recommendation dashboard

IR3 – Expiration Alerts	Users must quickly identify ingredients approaching expiration.	Color-coded warning indicators
IR4 – Grocery List Interface	Users must manage missing ingredients easily.	Interactive grocery checklist
IR5 – Analytics Dashboard	Users must review pantry usage analytics visually.	Charts and visualization dashboards
IR6 – Meal Planning Interface	Users must organize and review planned meals.	Calendar-style planning layout
IR7 – Inventory Management Screen	Users must add and update pantry items efficiently.	Dynamic inventory input forms
IR8 – Search and Filter Interface	Users must filter meals and pantry items easily.	Search bar and filtering controls
IR9 – Mobile Responsive Layout	Interface must remain usable across different screen sizes.	Responsive UI design principles
IR10 – Recommendation Explanation Display	Users must understand why recommendations were generated.	Recommendation explanation panels

Challenges

The most challenging requirements for SmartPantry are expected to include the following:

- Designing a recommendation engine that balances practicality, ingredient availability, expiration urgency, cooking time, and user preferences without generating repetitive recommendations.
- Managing pantry inventory efficiently while minimizing user frustration related to manual inventory entry and ingredient management.
- Matching pantry ingredients against recipe datasets accurately when ingredient naming conventions may differ between user inventory data and recipe database entries.
- Maintaining recommendation quality when users have limited pantry inventory or incomplete ingredient information available.
- Designing an interface that remains clean, understandable, and easy to navigate while supporting multiple connected systems such as recommendation generation, inventory management, grocery planning, and analytics dashboards.
- Preventing unnecessary scope expansion during development. Features such as barcode scanning, OCR receipt scanning, nutrition analysis, and collaborative household inventory management may significantly increase development complexity and are intentionally excluded from the MVP version of the project.
- Managing analytics processing and recommendation scoring efficiently while maintaining acceptable application performance and response times.
- Supporting scalable database organization for pantry history, recommendation history, grocery list generation, and user preference management.

One additional challenge involves recommendation explanation transparency. Users may lose trust in recommendation systems if they do not understand why certain meals were suggested. Because of this, SmartPantry will include recommendation explanation panels that display the primary recommendation factors used to generate meal rankings.

Research and Findings

Research on household food waste and meal planning applications strongly influenced the overall scope, features, and software architecture selected for SmartPantry. Competitor analysis revealed that many existing applications focus heavily on recipe discovery and meal inspiration while placing less emphasis on pantry awareness, expiration management, and recommendation practicality. Review analysis also showed that users become frustrated when pantry management systems require excessive manual input or when recipe recommendations become repetitive and unrealistic. These findings directly influenced the decision to prioritize simplified inventory management, expiration urgency analysis, recommendation practicality, and usability-focused interface design.

The multi-factor recommendation engine was selected as the primary system feature because it provides stronger recommendation flexibility than simple ingredient matching systems. Instead of recommending meals based only on ingredient overlap, SmartPantry combines ingredient availability, expiration urgency, missing ingredient count, cooking time, and user preferences into a weighted recommendation scoring model. This approach creates recommendations that better match real household situations. Python-based technologies such as Pandas, Flask or Streamlit, SQLite or PostgreSQL, and visualization libraries were selected because they support recommendation systems, analytics processing, database management, and dashboard development while remaining appropriate for a graduate-level software engineering capstone project.

SQLite and PostgreSQL were selected because they provide lightweight but scalable database solutions suitable for inventory management, recommendation history tracking, and analytics storage. Pandas was selected for analytics processing and recommendation calculations due to its strong support for structured data analysis and manipulation. The user stories and use cases were designed to reflect realistic household scenarios involving meal planning, grocery management, expiration tracking, and pantry organization. The overall system scope was intentionally structured to balance technical depth, recommendation-system research, usability testing, analytics integration, and realistic development expectations within a four-month capstone development cycle.

The final project combines recommendation systems, analytics, inventory management, human-computer interaction, and software engineering architecture into a unified application designed to address a real-world household problem.

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